

NAME: _____PRACTICE MIDTERM I

Please show your work and write only in **pen**. Notes are forbidden. Calculators, and all other electronic devices, are forbidden. Brains are encouraged, but at most one may be used per exam.

1: Please state Lagrange's Theorem correctly.

Up to minor variations in wording or choices of symbols, you should say something close to the following:

Lagrange's Theorem *Given any finite group G with subgroup H , $\#H \mid \#G$.*

Note 1: It appears that Lagrange merely discussed some special cases of the theorem we attribute to him: the first proof appears to have been one published by Pietro Abbati Marescotti in 1803.

Note 2: A slightly more high-brow version, which works for infinite groups too, is the following:

Given any group G with subgroup H , let $\{gH \mid g \in G\}$ denote the set of left cosets of H in G . Then $H \times \{gH \mid g \in G\}$ and G have the same cardinality.

2: (a) Please prove that if G is any **non**-cyclic group of cardinality 4, then there are exactly three elements of order 2.
 (b) Continuing (a), please show that if the three order 2 elements of G are α, β, γ , then $\alpha\beta=\gamma$. (Note that the order of the labelling doesn't matter!)

(a) Such a G clearly has 3 non-identity elements. By Lagrange's Theorem, their orders must be 2 or 4. Since G is not cyclic, there can be no order 4 elements and we are done.

(b) In essence, any equality *but* $\alpha\beta=\gamma$ would yield a contradiction. Here is what happens in each case:

(1) $\alpha\beta=e \implies \alpha=\beta^{-1} \implies \alpha=\beta$ (since $\beta^2=e \implies \beta=\beta^{-1}$), and this contradicts $\alpha \neq \beta$.

(2) $\alpha\beta=\alpha \implies \beta=e$ and this contradicts β having order 2.

(3) $\alpha\beta=\beta$ would contradict α having order 2, by an argument almost identical to that of (2).

So the only possibility remaining is $\alpha\beta=\gamma$. Note that it is conceivable that $\alpha\beta=\gamma$ could lead to a contradiction too, so we should at this point *construct* a group where such an identity holds. V (the Klein 4-group) is just such an example. In fact, we can even label the elements any way we like, provided α, β , and γ are assigned to distinct order 2 elements. ■

3: Someone (truthfully) tells you that G has an element a satisfying $a^{47}=e$. Someone else (truthfully) tells you that a also satisfies $a^{50}=e$. Prove that a must be the identity element.

The key here is to express 1 as an integer linear combination of 47 and 50, and then apply the identity to powers of a . To do the former, we can either guess cleverly, or just use the Extended Euclidean Algorithm. Going the latter route, we obtain

$$\begin{aligned} 50 &= 1 \cdot 47 + 3 \\ 47 &= 15 \cdot 3 + 2 \\ 3 &= 1 \cdot 2 + 1 \\ 2 &= 2 \cdot 1 + 0 \end{aligned}$$

and thus we need to invert the matrix

$$M := \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 15 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 50 & 17 \\ 47 & 16 \end{bmatrix}$$

The inverse is $\frac{1}{1} \begin{bmatrix} 16 & -17 \\ -47 & 50 \end{bmatrix}$ (and $\det M = 1$ since the number of matrices in our product above is even), so we at last obtain $16 \cdot 50 - 17 \cdot 47 = 1$.

To conclude, note then that $e = e \cdot e = e^{16}e^{-17} = (a^{50})^{16}(a^{47})^{-17} = a^{16 \cdot 50 - 17 \cdot 47} = a^1 = a$. ■

4: Please prove that all groups of cardinality ≤ 5 are Abelian.
[Done in class...](#)

5: Please prove that any group G with $g^2=e$ for all $g \in G$ must be Abelian.
Let $a, b \in G$ be any two elements of our group. By assumption, $a^2=e$. So then

$$a = a^{-1}, \tag{1}$$

by multiplying the previous identity by a^{-1} on the right. Similarly,

$$b = b^{-1} \tag{2}$$

(since $b^2=e$) and $ab=(ab)^{-1}=b^{-1}a^{-1}$ (since $(ab)^2=e$).

To conclude, we thus obtain $ab=b^{-1}a^{-1}=ba$ by applying (1) and (2). ■

6: Please prove that for any nonnegative integer k and a prime p , we have $\varphi(p^k) = p^k - p^{k-1}$.
Done in class.

7: Please prove that, for any distinct primes p and q , $\varphi(pq) = \varphi(p)\varphi(q)$.
Done in class.

8: Please prove Euler's Theorem: Given any integers a and m with $\gcd(a, m) = 1$,
 $a^{\varphi(m)} \equiv 1 \pmod{m}$.
[Done in class.](#)